

Radiological Activation and Associated Detriments of Samarium-Cobalt Permanent Magnets in High-Radiation Accelerator Environments

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1 Introduction to Permanent Magnet Technology in Particle Accelerators

The evolution of particle accelerator design over the past half-century has been heavily influenced by advances in magnetic materials. Early synchrotrons, cyclotrons, and linear accelerators relied almost exclusively on large, power-intensive, and water-cooled normal-conducting electromagnets to generate the dipole, quadrupole, and sextupole fields required for particle beam steering, focusing, and chromatic correction. While effective, these electromagnets impose massive infrastructural burdens, requiring extensive power cabling, highly purified cooling water systems, and substantial continuous electrical power consumption, which heavily inflates the facility's carbon footprint and operational costs.

The introduction of rare-earth permanent magnets (REPMs) in the mid-1960s and their maturation through the 1980s offered a paradigm-shifting alternative. Klaus Halbach's pioneering development of specialized magnet arrays, where the magnetic field is concentrated on one side of the array while being canceled on the other, unlocked the ability to construct extremely compact, high-gradient insertion devices, such as wigglers and undulators, entirely out of permanent magnetic materials. By utilizing REPMs, facility engineers could achieve peak fields comparable to conventional electromagnets while completely eliminating power supplies and cooling systems, thereby facilitating the "Zero Power Tuneable Optics" (ZEPTO) concepts pursued by modern facilities like the Compact Linear Collider (CLIC) and the European Synchrotron Radiation Facility (ESRF).

The two dominant families of high-strength rare-earth permanent magnets are Neodymium-Iron-Boron (NdFeB) and Samarium-Cobalt (SmCo). NdFeB represents the apex of magnetic strength, achieving maximum energy products ($BH_{\max}$) up to approximately 440 kJ/m³ for the highest-grade sintered materials, alongside superior remanent induction (B_r). Consequently, NdFeB is the default choice for a vast array of commercial and scientific applications. However, in the highly specialized domain of particle accelerators, the physical survival of the magnet in a mixed-field radiation environment introduces a fundamental engineering trade-off.

When exposed to the intense ionizing and hadronic radiation fields endemic to particle accelerators, specifically neutrons, protons, and high-energy bremsstrahlung photons, NdFeB exhibits a pronounced sensitivity to radiation-induced demagnetization. Conversely, SmCo alloys (specifically the SmCo₅ and Sm₂Co₁₇ stoichiometric phases) demonstrate an extraordinary

resilience to radiation damage, routinely surviving massive fluences that would completely erase the magnetic signature of an equivalent NdFeB block. This resilience makes SmCo a highly desirable engineering choice for target proximity zones, beam dumps, and tightly constrained vacuum undulators.

However, this magnetic survivability is accompanied by a significant operational challenge: the severe radiological activation of the SmCo matrix. The specific nuclear properties of Samarium and Cobalt result in the generation of intense, long-lived radioisotopes when exposed to continuous accelerator radiation. This technical note provides an analysis of the detriments associated with the radiological activation of SmCo magnets. By exploring the fundamental nuclear physics of transmutation, characterizing the specific isotopic hazards, and evaluating the effects on facility maintenance, dosimetry, and decommissioning, this document establishes a framework for evaluating the deployment of SmCo in high-fluence accelerator zones.

2 The Physics of Accelerator Radiation Environments

To accurately assess the detriment of SmCo activation, it is necessary to detail the specific radiation environment generated within a high-energy particle accelerator. Accelerators function by using resonant radiofrequency (RF) cavities to impart immense kinetic energy to charged particles (electrons, protons, or heavy ions) in a vacuum, utilizing magnetic fields to confine the beam into a stable trajectory.

Despite sophisticated beam loss monitoring and collimation systems, a fraction of the primary particle beam inevitably interacts with the physical infrastructure of the machine. This includes intentional interactions, such as the beam striking a spallation target to produce neutrons or entering a beam dump at the end of a run, as well as parasitic losses where beam halos scrape against vacuum chamber walls, drift tubes, or collimator jaws.

When a primary beam particle exceeding roughly 100 MeV strikes a high-Z (atomic number) material like copper, steel, or tungsten, it initiates a violent intra-nuclear cascade known as spallation. This reaction ejects secondary particles, including high-energy protons, charged pions, and notably, an intense flux of fast neutrons. Because neutrons lack an electric charge, they are unaffected by the magnetic fields guiding the primary beam and easily penetrate the thin walls of the vacuum beampipe, radiating isotropically into the surrounding accelerator enclosure.

As these fast neutrons interact with the concrete shielding, iron yokes, and surrounding earth, they undergo inelastic and elastic scattering, losing kinetic energy until they reach thermal equilibrium with their environment (approximately 0.025 eV). Consequently, permanent magnets installed near the beamline are bathed in a continuous, mixed-field radiation environment characterized by both high-energy spallation products and a dense, thermalized neutron gas.

In electron accelerators, such as synchrotron light sources or free-electron lasers (FELs), the dominant hazard mechanism shifts slightly. When high-energy electrons (e.g., the 2.5 GeV beam at the ESRF or the 180 MeV linac structures) stray from their orbits and impact matter, they generate intense bremsstrahlung (braking radiation) X-rays and gamma rays. If these photons possess energy exceeding the binding energy of the target nuclei (typically 8 to 15 MeV), they induce photonuclear reactions (γ, n), which subsequently eject neutrons into the surrounding environment. Regardless of whether the primary beam consists of protons or electrons, the resulting secondary neutron flux is the primary vector for the nuclear transmutation and subsequent radiological activation of nearby permanent magnets.

3 The CEBAF Specific Radiation Environment

While the preceding section outlines generalized mixed-field environments, assessing permanent magnet degradation requires defining the specific radiation parameters within the Continuous Electron Beam Accelerator Facility (CEBAF) enclosure. CEBAF operates a continuous-wave electron beam with energies currently reaching 12 GeV. In an electron machine of this energy, the primary vector for permanent magnet activation is not direct hadronic spallation, but rather photonuclear reactions driven by bremsstrahlung cascades.

When the primary electron beam interacts with physical structures such as septa, collimators, beam dumps, or residual gas within the vacuum chamber, it decelerates rapidly, emitting an intense forward-directed cone of bremsstrahlung photons. If these secondary photons possess energies that exceed the neutron binding energy of the surrounding atomic nuclei (typically peaking at the Giant Dipole Resonance between 15 MeV and 20 MeV for structural metals), they trigger (γ, n) photonuclear reactions in the copper cavities, stainless steel beampipes, and aluminum shielding [15].

This process ejects a dense flux of secondary fast neutrons. These neutrons scatter through the tunnel infrastructure, losing kinetic energy until they form a pervasive thermal and epithermal neutron gas. Consequently, any permanent magnets installed along the CEBAF beamline, even those shielded from direct electron impacts, are continuously subjected to this thermalized neutron flux, which serves as the primary catalyst for the isotopic transmutation and activation of the SmCo matrix.

4 The Trap of Magnetic Resilience: SmCo vs. NdFeB

To understand why facility engineers are tempted to use a material as radiologically problematic as SmCo, one must examine the specific mechanics of radiation-induced demagnetization.

Magnetic materials are characterized by their intrinsic coercivity (H_{ci}), which is a measure of the material’s resistance to becoming demagnetized by external magnetic fields, and their remanence (B_r), which dictates the strength of the magnetic field it produces. When exposed to radiation, it is the remanence that suffers the most visible degradation.

The primary mechanisms for this degradation are bulk displacement damage (point defect accumulation disrupting domain wall pinning sites) and localized “thermal spikes” in high-energy collision cascades. When an energetic particle, such as a fast neutron or a high-energy proton, collides with an atom within the magnet’s crystalline lattice, it transfers substantial kinetic energy, creating a Primary Knock-on Atom (PKA). This initiates a collision cascade, producing point defects and depositing localized heat. If the temperature within a localized domain exceeds the material’s Curie temperature (T_c), the magnetic alignment of that specific domain is violently randomized. Upon rapidly cooling, the domain remagnetizes, but often in an orientation dictated by the local demagnetizing field, effectively negating its contribution to the bulk magnetic output of the block.

Neodymium-Iron-Boron (NdFeB) magnets are exceptionally vulnerable to these phenomena. While NdFeB contains roughly 1% Boron by weight, the ^{10}B isotope (which constitutes $\sim 19.9\%$ of natural Boron) possesses an enormous cross-section (~ 3840 barns) for thermal neutron absorption, initiating a $^{10}\text{B}(n, \alpha)^7\text{Li}$ reaction. Both the resulting alpha particle and the ^7Li recoil nucleus deposit intense kinetic energy over extremely short ranges within the lattice, creating highly localized damage zones. This reaction, combined with extensive displacement damage from Iron and Neodymium PKA cascades, degrades the magnetic lattice. Furthermore, NdFeB possesses a relatively low Curie temperature of approximately 310°C to 370°C , rendering it highly susceptible to the thermal aspects of these cascades.

By stark contrast, Samarium-Cobalt magnets lack the vulnerable Boron isotope and feature an exceedingly high Curie temperature ranging from approximately 720°C for SmCo₅ to over 800°C for Sm₂Co₁₇. This high thermal ceiling provides immense thermodynamic headroom, allowing the SmCo matrix to absorb the energy of collision cascades without the local domains experiencing a phase transition.

Direct comparative testing validates this discrepancy. Experimental studies at Brookhaven National Laboratory evaluated both NdFeB and Sm₂Co₁₇ magnets (integrated within ferrofluidic feedthrough rotary seals relevant to FRIB applications) under exposure to mixed spallation irradiation fields consisting of neutrons, protons, and gamma rays. The NdFeB magnets exhibited substantial demagnetization at absorbed doses of 50 MRad, while the Sm₂Co₁₇ magnets remained essentially insensitive to irradiation up to a massive absorbed dose of 2 GRad [1]. Earlier studies at the Los Alamos National Laboratory (LANL) demonstrated that SmCo magnets exhibit far superior radiation hardness compared to NdFeB: at fast neutron fluences on the order of 10¹⁶ n/cm², SmCo samples showed negligible remanence loss, while NdFeB samples exhibited significant degradation that depended strongly on grade, coercivity, and irradiation temperature [2, 3].

Table 1: Comparative physical and magnetic properties of relevant accelerator magnet materials.

Magnet Type	Typical B_r (T)	Curie Temp. (T_c)	Thermal Coeff. of B_r	Radiation Resistance Profile
NdFeB (Sintered)	1.08–1.50	310–370°C	−0.11%/°C (grade dep.)	Poor. Demagnetization via displacement cascades and ¹⁰ B(n, α) reactions at fluences $\sim 10^{14}$ n/cm ² [2].
SmCo (Sm ₂ Co ₁₇)	0.95–1.15	800–850°C	−0.035%/°C	Excellent. Survives up to 2 GRad mixed-field absorbed dose [1] or $\geq 10^{18}$ n/cm ² [4].
Ferrite (Strontium)	0.38–0.40	$\sim 450^\circ\text{C}$	−0.20%/°C	Moderate to Good. Lower base field strength limits utility.

This extreme resilience establishes a compelling engineering incentive. Facility designers, seeking to avoid the operational downtime associated with frequently replacing and remagnetizing radiation-damaged NdFeB arrays, are highly motivated to deploy SmCo in the most hostile regions of the accelerator. However, the physical processes that govern this magnetic survival operate entirely independently from the nuclear processes that govern isotopic transmutation. While the SmCo domains do not flip, the constituent nuclei themselves act as massive sponges for the secondary radiation field, triggering significant radiological consequences.

5 The Physics of Transmutation and Radiological Activation in SmCo

Radiological activation occurs when stable isotopes absorb incident particles (typically neutrons) or undergo high-energy interactions (spallation, photodisintegration) that alter their nuclear mass or proton count, driving them into an unstable state. These unstable isotopes

then seek equilibrium by undergoing radioactive decay, emitting alpha particles, beta particles, gamma rays, or X-rays in the process.

The rate of activation for any specific element within an accelerator environment is governed by its nuclear cross-section, denoted in barns ($1 \text{ barn} = 10^{-28} \text{ m}^2$). A larger cross-section indicates a higher probability of interaction. Both Cobalt and Samarium feature isotopes with remarkably high cross-sections, resulting in rapid and intense activation profiles that drastically eclipse the background activation of surrounding structural materials like stainless steel or aluminum.

5.1 The Cobalt Hazard: The Generation of ^{60}Co and ^{58}Co

A pure stoichiometric $\text{Sm}_2\text{Co}_{17}$ phase consists of approximately 77% Cobalt by weight, while the SmCo_5 phase is roughly 66% Cobalt. However, commercial radiation-hardened 2:17 magnets, such as those evaluated for high-radiation environments, are typically heavily doped $\text{Sm}(\text{Co}, \text{Fe}, \text{Cu}, \text{Zr})_z$ alloys. This iron and copper doping dilutes the actual Cobalt mass fraction to approximately 50% [5]. Naturally occurring Cobalt is monoisotopic, consisting entirely of the stable isotope Cobalt-59 (^{59}Co).

When a SmCo magnet is placed in the secondary radiation field of an accelerator, the ^{59}Co nuclei are subjected to the ambient thermal and epithermal neutron flux. ^{59}Co has a substantial thermal neutron capture cross-section of approximately 37 barns [10]. Upon absorbing a neutron, it undergoes an (n, γ) radiative capture reaction, transmuting directly into Cobalt-60 (^{60}Co).

The radiological properties of ^{60}Co represent one of the most severe hazards in nuclear engineering and health physics:

1. **Half-Life:** ^{60}Co possesses a relatively long half-life of 5.27 years. This is short enough to yield a high specific activity (decays per second per gram), but long enough that it will not meaningfully decay away during standard facility operational timelines.
2. **Decay Modalities:** ^{60}Co decays via beta-minus (β^-) emission to an excited nuclear state of Nickel-60 (^{60}Ni). The excited ^{60}Ni nucleus instantaneously relaxes to its ground state by emitting a cascade of two highly energetic gamma rays, with precise energies of 1.17 MeV and 1.33 MeV.

The hazard posed by these gamma emissions cannot be overstated. Gamma rays at $\sim 1.25 \text{ MeV}$ possess immense penetrating power, capable of passing through air, biological tissue, and standard structural metals with minimal attenuation. While alpha and beta emissions can generally be mitigated by millimeters of plastic, aluminum, or standard personal protective equipment (PPE), attenuating a 1.25 MeV gamma field requires thick, high-Z (high atomic number) shielding, such as inches of solid lead or feet of high-density concrete. Because the commercial SmCo magnet is physically composed of approximately half Cobalt by weight, an irradiated SmCo array effectively transforms into a high-activity equivalent source.

While more significant in high-energy proton accelerator environments, the activation profile is further complicated by fast-neutron and direct proton spallation. High-energy incident particles can trigger $(n, 2n)$ or (p, pn) knockout reactions on the stable ^{59}Co matrix, generating Cobalt-58 (^{58}Co). The threshold for the $^{59}\text{Co}(n, 2n)^{58}\text{Co}$ reaction is approximately 10.5 MeV. Note that the secondary neutron spectrum from photonuclear reactions at CEBAF extends above this threshold, making this pathway relevant for electron machines as well. While ^{58}Co has a shorter half-life of 70.86 days, it decays via positron emission and electron capture, accompanied by the emission of a highly penetrating 0.811 MeV gamma ray. The presence of ^{58}Co exacerbates the immediate, acute residual dose rate of the magnet in the initial months following a beam shutdown, layering on top of the persistent ^{60}Co baseline.

5.2 The Samarium Hazard: Complex Decay Chains and Lanthanide Toxicity

Unlike Cobalt, naturally occurring Samarium is a complex mixture of seven isotopes: ^{144}Sm (3.1%), ^{147}Sm (15.0%), ^{148}Sm (11.2%), ^{149}Sm (13.8%), ^{150}Sm (7.4%), ^{152}Sm (26.7%), and ^{154}Sm (22.7%) [10]. While ^{147}Sm and ^{148}Sm are technically radioactive with extremely long half-lives (1.06×10^{11} years and 7×10^{15} years, respectively), they are effectively stable on any operational timescale. The neutron bombardment of this isotopic mixture triggers a diverse array of activation pathways, creating both intense short-term specific activities and permanent long-term liabilities.

The Short-Term Hazard: Samarium-153 (^{153}Sm)

The stable isotope ^{152}Sm possesses a significant thermal neutron activation cross-section of approximately 206 barns [10]. Through a standard (n, γ) reaction, it transmutes into Samarium-153 (^{153}Sm). ^{153}Sm has a half-life of 46.5 hours and decays via beta emission accompanied by a 103 keV diagnostic gamma ray [9]. While ^{153}Sm decays relatively quickly, its massive specific activity during the first few days of an accelerator shutdown produces localized radiation fields that severely limit immediate hands-on access to the undulator or magnetic lattice.

The Medium-Term Hazard: Samarium-145 (^{145}Sm)

The stable ^{144}Sm isotope undergoes neutron capture to produce Samarium-145 (^{145}Sm). With a thermal cross-section of approximately 1.6 barns [10], this is a minor activation pathway compared to ^{152}Sm , but it produces ^{145}Sm with a half-life of 340 days [6]. ^{145}Sm decays primarily via electron capture. While it lacks the high-energy gamma cascades of ^{60}Co , its near-one-year half-life ensures that the bulk magnetic material remains radiologically classified for years, complicating storage and handling protocols.

The Long-Term Hazard: Samarium-151 (^{151}Sm)

Perhaps the most insidious dosimetric hazard derived from Samarium is the production of Samarium-151 (^{151}Sm). Produced via the (n, γ) transmutation of the stable ^{150}Sm isotope, ^{151}Sm is primarily a beta emitter (with a negligible 21.5 keV gamma) with an extraordinarily long half-life. Published evaluations report values ranging from 90 ± 8 years [7] to the more recent NUBASE2020 adopted value of approximately 94.6 years [8].

Because of this ~ 90 -year half-life, the inventory of ^{151}Sm built up inside an irradiated SmCo magnet during an accelerator's operational lifespan will effectively *never* decay to background levels within the timeframe of human operations. While its predominantly low-energy beta emission profile means it does not contribute heavily to the external penetrating gamma dose rate (which is dominated by ^{60}Co), the presence of ^{151}Sm turns the magnet into a severe internal dosimetry hazard if the material is ever chipped, machined, or aerosolized.

Transmutation to Europium: The Gamma Escalation

The generation of potent, long-lived Europium gamma emitters occurs via two distinct pathways. First, the stable ^{154}Sm isotope (roughly 22.7% of natural Samarium, with a thermal cross-section of ~ 8.4 barns [10]) undergoes primary (n, γ) capture to produce ^{155}Sm . This isotope beta-decays rapidly (22.3-minute half-life) directly into Europium-155 (^{155}Eu , 4.76-year half-life).

Second, the beta-decay pathways of highly irradiated ^{151}Sm and ^{153}Sm terminate in stable Europium daughters (^{151}Eu and ^{153}Eu). Notably, ^{151}Eu has an immense thermal neutron capture cross-section of approximately 9,200 barns [10], making it a significant neutron poison and a major driver of subsequent activation. As the accelerator operates over decades, these newly formed stable daughters absorb ambient thermal neutrons via secondary (n, γ) captures. This sequential transmutation generates Europium-152 (^{152}Eu , 13.5-year half-life; dominant gammas at 344, 779, 964, and 1408 keV) and Europium-154 (^{154}Eu , 8.6-year half-life; dominant

gammas at 123, 723, 873, and 1275 keV) [9]. All three of these radioactive Europium isotopes establish a secondary, persistent high-intensity gamma field that parallels the hazard of ^{60}Co , further hardening the material’s unapproachable radiological profile.

Table 2: Principal activation pathways and radiological profiles of SmCo constituent elements in accelerator environments.

Parent	Reaction	Product	Half-Life	Emissions	Operational Impact
^{59}Co	(n, γ)	^{60}Co	5.27 yr	β^- ; γ : 1.17, 1.33 MeV	Severe external gamma dose; dominates long-term residual fields.
^{59}Co	$(n, 2n)$	^{58}Co	70.86 d	β^+/EC ; γ : 0.811 MeV	Exacerbates acute gamma dose rates post-shutdown.
^{154}Sm	(n, γ)	$^{155}\text{Eu}^*$	4.76 yr	β^- ; γ : 87, 105 keV	Long-term persistent gamma/beta hazard. (*via ^{155}Sm β^- decay.)
^{152}Sm	(n, γ)	^{153}Sm	46.5 hr	β^- ; γ : 103 keV	Intense short-term activity delays hands-on access.
^{150}Sm	(n, γ)	^{151}Sm	~ 90 yr	β^- (low energy)	Permanent waste liability; internal dosimetry hazard.
^{144}Sm	(n, γ)	^{145}Sm	340 d	Electron Capture	Broadly elevates medium-term background activity.

5.3 The ^{149}Sm Neutron Poison Effect and Material Burn-up

While the preceding sections outlined the radiological hazards to personnel, there is a secondary mechanism by which radiation damages the SmCo magnet itself: elemental burn-up.

Samarium contains the stable isotope Samarium-149 (^{149}Sm). In nuclear physics, ^{149}Sm is renowned as one of the most powerful elemental neutron poisons in existence. Its thermal neutron absorption cross-section of approximately 40,140 barns represents an extremely high interaction probability, leading to progressive elemental depletion in sustained neutron fields [10].

When a SmCo magnet is placed in a high-intensity thermal neutron field, the ^{149}Sm acts as a profound neutron sink. It absorbs thermal neutrons, transmuting into the stable ^{150}Sm . Driven by a cross-section of approximately 100 barns [10], this ^{150}Sm then undergoes a steady, secondary (n, γ) capture to become the long-lived radioactive ^{151}Sm .

In extreme high-fluence regimes, this continuous transmutation alters the elemental stoichiometry of the magnet’s physical lattice. Because permanent magnetism is fundamentally a structural property governed by precise crystallographic anisotropy, the volumetric transmutation of Samarium atoms physically degrades the metallurgical matrix. This elemental depletion is a direct consequence of the massive thermal neutron capture cross-sections detailed in evaluated nuclear data [10]. At the thermal neutron fluences encountered in high-loss accelerator zones, this depletion becomes non-negligible over multi-decade operational timescales. Thus, while SmCo is famously resistant to the localized *thermal spikes* that immediately demagnetize NdFeB, continuous exposure to immense neutron fluences eventually results in irreversible, structural magnetic degradation simply because the elemental Samarium ceases

to exist, having been transmuted into non-magnetic or differently-coordinated lanthanide daughters.

6 Real-World Facility Case Studies and Operational Detriments

The isotopic activation profile detailed above translates directly into severe, practical operational and logistical detriments for any accelerator facility utilizing SmCo components. The fundamental dichotomy between radiation *hardness* and radiological *activation* is repeatedly evidenced in real-world accelerator operations.

6.1 Diminished Efficacy of Cooling Times

In standard accelerator health physics, a core mitigation strategy for managing induced radioactivity is the enforcement of “cooling time.” Following a beam shutdown, access to the accelerator tunnel is strictly prohibited for a calculated duration (ranging from a few hours to several weeks). This temporal buffer allows short-lived activation products within the aluminum vacuum chambers, copper RF cavities, and steel yokes to naturally decay to baseline safe levels, permitting technicians to enter the vault and perform hands-on maintenance. Radiation protection protocols, such as ALARA (As Low As Reasonably Achievable), rely heavily on this predictable decay curve.

The presence of irradiated SmCo magnets significantly diminishes the efficacy of the cooling time strategy for localized zones. Because the dominant residual dose rate from the magnet assembly is driven by ^{60}Co , which has a half-life of 5.27 years, shutting down the accelerator for typical maintenance windows results in a negligible reduction in the ambient gamma field.

At the **CERN Antiproton Decelerator (AD)**, new radiation-hard SmCo permanent focusing magnets were installed directly in the high-power target facility [11]. Because the target area expects massive proton fluxes on the target material, extreme radiation protection measures were mandatory. The target and shielding had to be housed in a primary vacuum tank to reduce air activation, and the permanent magnets demand stringent remote handling protocols, as direct human intervention in the resulting activated environment is not operationally feasible.

Conversely, the designers of the **Cornell-BNL ERL Test Accelerator (CBETA)** evaluated the use of SmCo versus NdFeB for their permanent magnet quadrupoles. While acknowledging SmCo’s superior radiation resistance, the collaboration ultimately constructed the magnets out of NdFeB. The decision was driven directly by the radiological detriment: SmCo contains much more Cobalt, which leads to long-term ^{60}Co activation [12].

6.2 Local Maintenance Logistics and Dosimetry Planning

When remote robotics are unavailable and hands-on maintenance is strictly required near an irradiated SmCo array, health physics teams must rely on aggressive time, distance, and shielding protocols. However, the 1.17 and 1.33 MeV gamma emissions from ^{60}Co deeply complicate the use of local temporary shielding (often referred to as “shadow shielding”).

Attenuating these high-energy gammas by a single tenth-value layer (TVL) (reducing the dose rate by a factor of ten) requires approximately 2 inches of solid lead or nearly 4 inches of steel [16]. Wrapping or surrounding a tightly constrained vacuum chamber and undulator array with hundreds of pounds of lead is often physically impossible due to spatial and weight-bearing restrictions in the accelerator tunnel.

Consequently, routine maintenance on adjacent, un-activated systems (such as vacuum pumps, water cooling manifolds, or diagnostics) requires strictly monitored “stay times.” Technicians may only have minutes of allowed work time near the SmCo magnets before reaching their shift dose limits. This necessitates extensive pre-planning, mock-up training outside the vault, and rotating multiple personnel through a single task, which reduces facility operational efficiency and inflates the collective dosimetric burden.

6.3 The Brittleness Paradox: Physical Integrity and Contamination Hazards

The challenges of working around these structures are compounded significantly by the intrinsic physical properties of rare-earth permanent magnets. SmCo, being an intermetallic compound, is notoriously brittle and ceramic-like in its mechanical behavior. It possesses exceptionally low fracture toughness and is highly susceptible to chipping, spalling, and catastrophic fracture upon experiencing mechanical shock, excessive clamping force, or rapid thermal gradients [13]. To mitigate this, SmCo magnets are often encased in electrolytic nickel surface plating or aluminum holders, but the underlying matrix remains fragile.

This inherent brittleness compounds the hazards associated with the magnet’s radiological activation status. When a technician or telemanipulator arm attempts to maneuver near or move a highly radioactive SmCo block, the operator must fight against the powerful attractive or repulsive remanent magnetic forces acting upon steel tooling or adjacent beamline infrastructure. If the magnet accidentally snaps against a steel rail or is compressed too tightly by a wrench or robotic gripper, the SmCo block can easily crack or shatter.

When an activated SmCo magnet fractures, the hazard profile instantaneously evolves from a purely external penetrating gamma threat (^{60}Co) to a severe internal contamination threat. The shattering of the lattice creates highly dispersible, radioactive micro-shrapnel and dust. If this dust contains the long-lived beta-emitting ^{151}Sm or the highly active ^{153}Sm , it presents a significant internal dosimetry hazard. These radio-lanthanides, if internalized, bioaccumulate and continuously irradiate surrounding tissues, constituting a significant internal contamination event. Consequently, any physical degradation of an activated SmCo magnet forces the facility into a full radiological emergency posture, requiring the deployment of rigorous contamination control protocols and comprehensive surface decontamination of the accelerator vault.

7 Decommissioning, Waste Management, and Environmental Impact

The end-of-life phase for particle accelerators employing SmCo magnetic architecture reveals the most enduring and expensive detriments. Decommissioning an accelerator facility involves the meticulous radiological characterization, physical dismantling, protective packaging, and ultimate geological disposal of all activated structural components.

7.1 Regulatory Waste Classification and Disposal Economics

In the United States, as well as under International Atomic Energy Agency (IAEA) frameworks, radioactive waste is strictly categorized based on its origin, its specific activity (measured in Bq/g or Ci/g), and crucially, the half-lives of its constituent isotopes. These classifications (such as Low-Level Waste Classes A, B, and C, or Greater-Than-Class-C) legally dictate the

required shielding configurations and the specific geological repositories permitted to receive the material [14].

Because irradiated SmCo magnets accumulate massive, sustained inventories of ^{60}Co (5.27 years), ^{152}Eu (13.5 years), and ^{151}Sm (~ 90 years), they can exceed the regulatory thresholds for standard, inexpensive shallow-land burial, particularly in high-fluence zones. The presence of the long-lived ^{151}Sm is particularly punitive from a regulatory standpoint. Environmental protection frameworks strictly limit the specific activity of such long-lived isotopes, should they leach from comminuted or corroded waste in near-surface disposal trenches due to the modeled risks of groundwater migration over century-long timescales.

Therefore, activated SmCo requires highly specialized, deeply shielded, corrosion-resistant waste containers and must often be shipped to dedicated, deeply geological federal repositories. The total financial burden of assaying, packaging, transporting, and perpetually storing this highly active waste can significantly exceed the initial procurement cost of the magnet assembly.

7.2 Complexity of Dismantling Highly Magnetic Rad-Waste

Decommissioning SmCo is not merely an administrative or shielding challenge; the intrinsic permanent magnetism of the material severely complicates the physical mechanics of disposal. Standard techniques for volume-reducing nuclear waste include plasma cutting, shearing, or crushing. Applying aggressive mechanical techniques like shearing or crushing to highly magnetic, brittle SmCo blocks is exceptionally hazardous. While thermal techniques like plasma cutting can heat the material above its Curie temperature ($\sim 800^\circ\text{C}$ for $\text{Sm}_2\text{Co}_{17}$) and beneficially demagnetize it, they simultaneously generate highly radioactive fume, swarf, and metallic vapor.

Furthermore, during mechanical handling, the intense, undiminished remanent fields of the bulk blocks cause them to violently attract to one another or to the steel walls of the radioactive waste drums. This creates severe pinch hazards for remote manipulators and makes dense volumetric packing of the waste drums nearly impossible without the fabrication of carefully engineered, non-magnetic internal spacers and jigs.

8 Comparative Radiologic Viability: SmCo vs. NdFeB

To fully contextualize the detriments of SmCo, it is necessary to contrast it with the primary alternative in the REPM market: Neodymium-Iron-Boron (NdFeB). The decision of which alloy to employ in an accelerator environment constitutes a fundamental engineering trade-off between absolute magnetic stability and lifecycle radiological safety.

8.1 The Radiation Hardness Discrepancy

As established, NdFeB is notoriously sensitive to radiation-induced demagnetization. Exposure to secondary electrons, bremsstrahlung, or low-fluence neutrons causes local domain reversals in NdFeB, leading to unacceptable flux losses. The mechanism involves $^{10}\text{B}(n, \alpha)^7\text{Li}$ reactions and bulk displacement damage, which deposit sufficient localized energy to exceed the material's relatively low Curie temperature ($\sim 310^\circ\text{C}$) within collision cascades. Conversely, SmCo demonstrates exceptional resistance to demagnetization, maintaining strict field tolerances up to extreme fluences [1].

8.2 The Radiological Profile Discrepancy

However, from a health physics and environmental detriment perspective, NdFeB is vastly superior to SmCo.

- **NdFeB Activation:** NdFeB contains roughly 65–70% Iron by weight. The primary activation products of iron in a typical accelerator environment are Manganese-54 (^{54}Mn , 312-day half-life) and Iron-55 (^{55}Fe , 2.7-year half-life, which primarily decays via electron capture with very weak, easily shielded X-ray emission). While Neodymium and Dysprosium (a common dopant) do activate, their resulting isotopes generally lack the intense, highly penetrating, and long-lasting gamma fields characteristic of Cobalt-60.
- **SmCo Activation:** As detailed extensively, the combination of Cobalt and Samarium yields an isotopic profile of 5.27-year, highly penetrating gammas (^{60}Co) and ~ 90 -year internal dosimetry hazards (^{151}Sm).

The comparative residual dose rate of an irradiated NdFeB magnet drops off considerably faster than that of an equivalent SmCo magnet. While an irradiated NdFeB undulator array might become safe for temporary hands-on maintenance after several months of cooling (as the ^{54}Mn decays), a SmCo array receiving the exact same primary fluence remains active for decades.

Table 3: Comparative lifecycle and operational impacts of NdFeB vs. SmCo magnets in accelerators.

Evaluation Metric	NdFeB	SmCo
Primary activation products	^{54}Mn , ^{55}Fe , ^{147}Nd	^{60}Co , ^{58}Co , ^{155}Eu , ^{153}Sm , ^{151}Sm
Dominant long-term hazard	Moderate (^{54}Mn : 312-day half-life gammas)	Severe (^{60}Co : 5.27-yr gammas; ^{151}Sm : ~ 90 -yr betas)
Cooling time efficacy	Highly effective over weeks to months	Diminished; requires years to decades
Hands-on maintenance	Possible after cooling period	Prohibited in high-fluence zones; requires robotics or strict stay-time limits
Waste classification	Generally lower-level waste	Often elevated to high-activity legacy waste
Magnetic radiation resistance	Poor (vulnerable to thermal spikes and ^{10}B reactions)	Excellent (stable up to 2 GRad fluences) [1]

8.3 The Modern Engineering Consensus and Mitigation Strategies

Because of the long-term detriments of radiologically activated SmCo, namely, the severe constraints on hands-on access to the accelerator tunnel and the decommissioning costs, modern accelerator design increasingly favors NdFeB, despite its poor radiation hardness.

Facilities will deliberately accept the engineering burden of designing heavy, localized radiation shielding specifically to protect the fragile NdFeB magnets from the secondary particle spray. For example, rather than using SmCo, modern free-electron lasers and light

sources often plan for the routine, scheduled replacement and remagnetization of radiation-damaged NdFeB components, viewing this recurring maintenance cost as vastly preferable to the permanent radiological liability of installing SmCo. The use of Praseodymium-Iron-Boron (PrFeB) is also being explored for extreme cryogenic conditions (down to 4 K) to avoid the spin reorientation phenomena seen in NdFeB, entirely bypassing the need for SmCo. Consequently, SmCo is now generally restricted by health physics committees to highly specific, unavoidable applications where heavy shielding is impossible due to strict physical aperture limits, or where extreme localized operating temperatures explicitly mandate its high Curie temperature.

9 Implications for the Proposed FFA-Based Energy Upgrade

The application of these radiological principles is highly relevant to future facility expansions, particularly the proposed Fixed-Field Alternating gradient (FFA) architecture for a potential CEBAF energy upgrade [17]. The FFA concept relies heavily on complex, strong-focusing multipole magnetic fields to transport multiple beam passes through a single arc. Halbach arrays utilizing permanent magnets are an ideal engineering solution for this architecture, offering the necessary field gradients without the massive power consumption and cooling infrastructure required by conventional electromagnets.

While CEBAF does not currently utilize FFA arcs, the design phase for this upgrade necessitates a critical choice between NdFeB and SmCo magnetic materials. Given the immense physical scale of the proposed FFA arcs, utilizing SmCo would require installing vast quantities of Cobalt and Samarium along the beamline.

Over the multi-decade operational lifetime of the upgraded accelerator, the distributed, unavoidable fractional beam losses associated with multi-pass transport will generate a continuous photonuclear neutron flux. If the FFA Halbach arrays are constructed from SmCo, this extensive magnetic lattice will globally activate. The entire arc would gradually transform into a distributed ^{60}Co , ^{152}Eu , and ^{154}Eu gamma source.

This presents an unmanageable ALARA challenge. Routine hands-on maintenance for adjacent beamline components, such as vacuum ion pumps, beam position monitors (BPMs), and diagnostic equipment, would incur substantial personnel dose simply due to their proximity to the activated SmCo arrays. Furthermore, at the facility's end-of-life, the dismantling and disposal of kilometers of highly activated, highly magnetic SmCo blocks would incur significant decommissioning costs. Therefore, despite the susceptibility of NdFeB to radiation-induced demagnetization, the consensus for large-scale installations like the proposed FFA upgrade heavily favors deploying NdFeB combined with strategic, localized radiation shielding to protect the magnets, thereby avoiding the permanent dosimetric liability of a SmCo lattice.

10 Conclusion

The utilization of Samarium-Cobalt (SmCo) permanent magnets in modern particle accelerators presents a severe and intractable engineering dichotomy. From a purely physical and operational standpoint, the material is structurally and magnetically superior, offering robust resistance to radiation-induced demagnetization, supreme thermal stability, and the elimination of massive power infrastructure. However, the radiological detriments exacted by the material decisively undermine its long-term viability in unshielded, high-fluence environments.

The high natural abundance and massive neutron absorption cross-sections of Cobalt-59 and Samarium isotopes lead to immediate, intense, and sustained nuclear activation. The

production of Cobalt-60 creates an intense, deeply penetrating gamma-ray field that effectively diminishes the utility of standard personnel “cooling times,” prohibiting extended hands-on maintenance in high-loss zones and forcing facilities to rely on shadow shielding, restrictive stay times, or complex remote-handling robotic methodologies. Furthermore, the generation of Samarium-151 and highly active Europium isotopes introduces severe internal dosimetry hazards, a threat profoundly compounded by the extreme mechanical brittleness of the alloy, which makes the material highly susceptible to fracturing and creating highly mobile radioactive contamination. The phenomenon of Samarium-149 “burn-up” under massive thermal neutron fluences can eventually cause even this robust material to suffer structural failure at extreme doses.

Real-world deployments at BNL/FRIB, CERN, and Cornell highlight this engineering trade-off: while SmCo robustly survives intense radiation environments, the resulting activation complicates hands-on maintenance. Ultimately, the lifecycle of a SmCo magnet in an accelerator terminates in a complex waste management liability. The generation of long-lived, high-activity isotopes forces facilities to weigh the magnetic stability gained during operation against diminished operational flexibility, elevated personnel exposure, and heightened geological disposal costs.

References

- [1] N. Simos, S. Ozaki, N. Mokhov, A. Zeller, W. Mittig, and F. Pellemoine, “Demagnetization of $\text{Nd}_2\text{Fe}_{14}\text{B}$, $\text{Pr}_2\text{Fe}_{14}\text{B}$, and $\text{Sm}_2\text{Co}_{17}$ Permanent Magnets in Spallation Irradiation Fields,” *IEEE Trans. on Magnetics*, vol. 54, no. 5, pp. 1–6, 2018. <https://doi.org/10.1109/TMAG.2017.2769040>
- [2] J. R. Cost, R. D. Brown, A. L. Giorgi, and J. T. Stanley, “Effects of Neutron Irradiation on Nd-Fe-B Magnetic Properties,” *IEEE Trans. on Magnetics*, vol. 24, no. 3, pp. 2016–2019, 1988. <https://doi.org/10.1109/20.11665>
- [3] D. B. Barlow, R. H. Kraus, and M. J. Borden, “Radiation Hardness Measurements of New Permanent Magnet Materials for High-Intensity Linac Applications,” *Los Alamos National Laboratory*, Rep. LA-UR-97-3370, 1997. <https://doi.org/10.2172/562305>
- [4] C. H. Chen et al., “The effect of neutron irradiation on Nd-Fe-B and $\text{Sm}_2\text{Co}_{17}$ -based high temperature magnets,” *IEEE Trans. on Magnetics*, vol. 41, pp. 3832–3834, 2005. <https://doi.org/10.1109/TMAG.2005.854774>
- [5] O. Gutfleisch et al., “Magnetic Materials and Devices for the 21st Century: Stronger, Lighter, and More Energy Efficient,” *Adv. Mater.*, vol. 23, pp. 821–842, 2011. <https://doi.org/10.1002/adma.201002180>
- [6] E. Browne and J. K. Tuli, “Nuclear Data Sheets for A=145,” *Nuclear Data Sheets*, vol. 113, pp. 715–908, 2012. <https://doi.org/10.1016/j.nds.2012.04.001>
- [7] M.-M. Bé et al., “Determination of the ^{151}Sm half-life,” *Applied Radiation and Isotopes*, vol. 68, no. 7–8, pp. 1398–1401, 2010. <https://doi.org/10.1016/j.apradiso.2010.01.031>
- [8] F. G. Kondev, M. Wang, W. J. Huang, S. Naimi, and G. Audi, “The NUBASE2020 evaluation of nuclear properties,” *Chinese Physics C*, vol. 45, no. 3, p. 030001, 2021. <https://doi.org/10.1088/1674-1137/abddae>

- [9] National Nuclear Data Center, “Evaluated Nuclear Structure Data File (ENSDF),” *Brookhaven National Laboratory*. <https://www.nndc.bnl.gov/ensdf/>
- [10] Evaluated Nuclear Data File (ENDF/B-VIII.0), *National Nuclear Data Center*, Brookhaven National Laboratory, 2018. <https://www.nndc.bnl.gov/ensdf/b8.0/>
- [11] C. Torregrosa et al., “CERN Antiproton Target Area Upgrade,” *Proc. of HB2023*, Geneva, Switzerland, 2023. <https://doi.org/10.18429/JACoW-HB2023-TUPM065>
- [12] S. Brooks et al., “Permanent Magnets for the Return Loop of the Cornell-BNL ERL Test Accelerator,” *Phys. Rev. Accel. Beams*, vol. 23, p. 112401, 2020. <https://doi.org/10.1103/PhysRevAccelBeams.23.112401>
- [13] A. Samin and M. Kuranz, “A Review of Radiation-Induced Demagnetization of Permanent Magnets,” *J. Nucl. Mater.*, vol. 503, pp. 42–55, 2018. <https://doi.org/10.1016/j.jnucmat.2018.02.016>
- [14] U.S. Nuclear Regulatory Commission, “Licensing Requirements for Land Disposal of Radioactive Waste,” *Title 10, Code of Federal Regulations, Part 61*, 2024. <https://www.nrc.gov/reading-rm/doc-collections/cfr/part061/>
- [15] P. Degtiarenko and G. Kharashvili, “Radiation Damage and Activation of Materials in the CEBAF 12 GeV Accelerator Environment,” *Jefferson Lab RadCon*, Newport News, VA, Tech. Note JLAB-TN-15-024, 2015. https://misportal.jlab.org/ul/publications/view_pub.cfm?pub_id=14187
- [16] H. Cember and T. E. Johnson, *Introduction to Health Physics*, 4th ed., McGraw-Hill, 2008. ISBN: 978-0071462676.
- [17] K. E. Deitrick et al., “CEBAF Energy Upgrade using FFA Arcs,” *Proc. of IPAC2023*, Venice, Italy, 2023. <https://doi.org/10.18429/JACoW-IPAC2023-WEPA088>